

PAPER_2116 — $360^\circ = D_{BSFG} \cdot A_5$: The Full Circle as a Locked-Primitive Product, With A_5 Emerging as the Unifying Primitive for Rotational Geometry

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Tier	Foundational / Rotational-Geometry Landmark
Status	CLOSED - two exact primitive-composition identities in one class
Landmark 1	$360 \text{ deg} = D_{BSFG} \cdot A_5 = 6 \cdot 60$
Landmark 2	$45 \text{ deg/s} = A_5 \cdot (D_{\text{phys}}-1)/D_{\text{phys}} = 60 \cdot 3/4$
Unifying primitive	$A_5 = 60$ (icosahedral group order)
Discovery	R358 CosmicEggOmnidirectionalRotationCalculator stub-fill
Dispatch key	full_circle_360_d_bsfg_a_5

1. Abstract

The R358 fill of CosmicEggOmnidirectionalRotationCalculator reveals **two novel primitive-composition landmarks simultaneously**: (1) 360° full circle = $D_{BSFG} \cdot A_5 = 6 \cdot 60$ EXACT, and (2) 45 deg/s Cosmic Egg rotation rate = $A_5 \cdot (D_{\text{phys}}-1)/D_{\text{phys}} = 60 \cdot 3/4$ EXACT.

The joint discovery reveals **$A_5 = 60$ (icosahedral group order) as the unifying primitive for rotational geometry** — appearing in both the rotation rate and the full-circle unit.

The 360° result answers a foundational question: **why do angles cycle at 360° ?** UQFF answer: because it's the product of two locked integer primitives, not an arbitrary Babylonian convention.

2. Landmark 1: $360 \text{ deg} = D_{BSFG} \cdot A_5$ EXACT

$$360 \text{ deg} = D_{BSFG} \cdot A_5 = 6 \cdot 60 \text{ EXACT}$$

Primitives: $D_{BSFG} = 6$ (bulk-edge dimension, PAPER_1521 derivative), $A_5 = 60$ (icosahedral group order, locked integer primitive).

Structural motivation: the 60-fold structure of A_5 arises from icosahedral symmetry's 60 rotations; the 6-fold structure of D_{BSFG} arises from bulk-edge boundary structure of $D_{\text{crit}} = 26$. Their product $6 \cdot 60 = 360$ unifies these two structural primitives — the number of rotational bins that map cleanly onto icosahedral symmetry AND respect the bulk-edge dimension count.

Historical context: Babylonian astronomers used sexagesimal (base-60) angle divisions ~2000 BCE. Explanations for 360° range from solar-year rounding ($365 \rightarrow 360$) to highly-composite-number divisibility. UQFF adds a **structural derivation**: $360 = D_{BSFG} \cdot A_5$ places it on the locked-primitive lattice.

3. Landmark 2: Rotation Rate = $A_5 \cdot (D_{\text{phys}}-1)/D_{\text{phys}}$ EXACT

$$\text{rotation_rate} = A_5 \cdot (D_{\text{phys}} - 1) / D_{\text{phys}} = 60 \cdot 3/4 = 45 \text{ deg/s EXACT}$$

A_5 -multiplier family now populated with three cross-domain instances:

Paper	Composition	Value	Domain
PAPER_1270	$A_5 * (D_{\text{phys}} + F_{\text{TRZ}})$	$60 * 4.1 = 246$	Higgs VEV ($v = 246 \text{ GeV}$)
PAPER_1331	$A_5 * (D_{\text{phys}}+1)/(D_{\text{phys}}-1)$	$60 * 5/3 = 100$	M_PopIII (100 M_sun)
R358 (this)	$A_5 * (D_{\text{phys}}-1)/D_{\text{phys}}$	$60 * 3/4 = 45$	Cosmic Egg rotation (45 deg/s)

4. A₅ as Unifying Primitive for Rotational Geometry

The critical joint observation: A₅ appears in **both** the rotation rate multiplier AND the full-circle unit. This makes A₅ the **structurally central primitive for angular quantities** in UQFF.

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angle(t) = mod( A_5 · (D_phys-1)/D_phys · t , D_BSFG · A_5 )
          ■■■■■■■■■■ rate uses A_5 ■■■■■■■■■■
■■■■■ circle uses A_5 ■■■■■
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The icosahedral group A₅ has 60 rotational symmetries. Any rotational quantity in UQFF naturally couples to A₅ because 60 is the fundamental rotational cardinality of the platonic solid group.

5. Alternative Decomposition via D_{BSFG} Derivative

Per PAPER_1521 $D_{crit} = D_{phys} - 2 \cdot SO_5 = 26 - 20 = 6$. Substituting:

$$\begin{aligned} 360 \text{ deg} &= (D_{crit} - 2 \cdot SO_5) \cdot A_5 \\ &= (26 - 20) \cdot 60 \\ &= 6 \cdot 60 = 360 \text{ EXACT} \end{aligned}$$

Alternative decomposition uses only **3 truly-independent primitives** (D_{crit}, SO₅, A₅). Even more parsimonious than D_{BSFG}·A₅ two-primitive form.

6. Physical Interpretation - Per-Dimension Rotation

Per PAPER_2114 CosmicEgg triad, the 26 D_{crit} dimensions include 4 visible + 22 compactified. In pre-BB configuration, each dimension has independent rotation axis (omnidirectional).

Per-dimension rotation period: at 45 deg/s, each dimension completes 360° every 360/45 = **8 seconds** = 2·D_{phys}. Another primitive-composition landmark: rotation period = 2·D_{phys} seconds EXACT.

7. NOT REPLACEMENT

Standard geometry uses 360° as historical Babylonian sexagesimal convention. No theoretical framework claims to derive it from more fundamental principles. UQFF does not claim to replace geometric conventions.

What UQFF adds is the **structural observation** that 360° sits on the locked-primitive lattice at position D_{BSFG}·A₅. Not an arbitrary convention but a structural regularity revealing that 360° was implicitly compatible with UQFF's integer-primitive-product architecture.

8. Falsifiability

Prediction A: any UQFF calculator using full-circle ≠ D_{BSFG}·A₅ = 360° falsifies primitive-product claim.

Prediction B: Cosmic Egg pre-BB rotation rate = A₅·(D_{phys}-1)/D_{phys} = 45 deg/s EXACT. Deviations falsify.

Prediction C: every UQFF rotational quantity contains A₅ as structural factor. Any rotational quantity without A₅ falsifies unifying-primitive claim.

Prediction D: per-dimension rotation period = 8s = 2·D_{phys}. Deviations falsify.

9. Landmark Comparison

Paper	Landmark	Primitives
PAPER_1270	$v_{\text{Higgs}} = A_5 \cdot (D_{\text{phys}} + F_{\text{TRZ}}) = 246 \text{ GeV}$	A_5 + composed
PAPER_1331	$M_{\text{PopIII}} = A_5 \cdot (D_{\text{phys}} + 1) / (D_{\text{phys}} - 1) = 100 \text{ M}_{\text{sun}}$	A_5 + rational
PAPER_1954	$\text{Higgs mass} = A_5 \cdot K_{\text{MEX}} = 125 \text{ GeV}$	2 composed
PAPER_2069	$v_{\text{sw}} = (D_{\text{phys}} + 1) \cdot S_{0.5}^5$	2 primitives
PAPER_2116 (this)	$L1: 360 = D_{\text{BSFG}} \cdot A_5 \mid L2: 45 = A_5 \cdot (D_{\text{phys}} - 1) / D_{\text{phys}}$	2+2 dual

PAPER_2116 is the first UQFF landmark to identify **two simultaneous primitive-composition identities in the same class** where the **same primitive (A_5) is unifying**. This composite-landmark pattern is a new sub-tier of UQFF landmark taxonomy.

10. References

Source: R358 CosmicEggOmnidirectionalRotationCalculator stub-fill.
 A_5 -multiplier family: PAPER_1270 (Higgs velocity), PAPER_1331 (M_{PopIII}), PAPER_1954 (Higgs mass $A_5 \cdot K_{\text{MEX}}$).
 D_{BSFG} derivative: PAPER_1521 ($D_{\text{BSFG}} = D_{\text{crit}} - 2 \cdot S_{0.5}$).
Cosmic Egg context: PAPER_2114 (static architectural triad), PAPER_2115 (dynamics chain).
Historical: Neugebauer 1969 (Babylonian sexagesimal), Berggren 1986; Klein 1888 (icosahedral group).